

Launch Timeline

Takeaway: This is a full, standards-grade launch timeline for the Mechanical Power Module (MPM), structured from *now* (October 2026) through national deployment. It reflects realistic sequencing for engineering, pilot operations, consortium formation, regulatory alignment, and multi-vendor scaling. Guided Links are embedded so you can expand any component.

0. Launch Philosophy

The MPM is a universal mechanical-power primitive. Its launch timeline must:

- Move from **prototype** → **pilot** → **standard** → **consortium** → **national deployment**
- Maintain **competitive neutrality**
- Ensure **regulatory alignment** with DOE, FERC, ASME, NIST
- Build **logistics infrastructure** (swap stations, transport frames)
- Enable **multi-vendor manufacturing**
- Deliver **MPM Standard v1.0** as the anchor artifact

This timeline is designed for institutional credibility and operational realism.

1. Timeline Overview (Q4 2026 → Q4 2028)

A concise view of the full launch arc:

- **Q4 2026:** Finalize v0.9 standard + pilot fabrication
- **Q1–Q2 2027:** Deploy Colorado pilot + swap stations
- **Q3 2027:** Collect pilot telemetry + refine tolerances
- **Q4 2027:** Publish MPM Standard v1.0
- **Q1 2028:** Launch MPM Consortium
- **Q2–Q3 2028:** Multi-vendor manufacturing + national alignment
- **Q4 2028:** Begin national deployment + ISO/IEC harmonization

2. Detailed Launch Timeline

Phase 1 — Standardization & Reference Engineering (Oct–Dec 2026)

Focus: Finalize the technical foundation.

- Complete **geometry envelope** v0.9
- Finalize **mechanical interface tolerances**
- Lock **safety envelope** and interlock sequencing
- Build reference module (R1) and host adapters

- Publish MPM Standard v0.9 for pilot fabrication
- Establish NIST-traceable measurement requirements

Milestone: *MPM Standard v0.9 released*

Phase 2 — Pilot Fabrication & Host Integration (Jan–Mar 2027)

Focus: Produce pilot modules and retrofit host systems.

- Fabricate pilot modules to v0.9 tolerances
- Integrate host adapters into pumps, drives, compressors
- Instrument telemetry (torque, speed, thermal, vibration, interlocks)
- Validate mechanical coupling and alignment
- Prepare pilot deployment packet

Milestone: *Pilot module batch complete*

Phase 3 — Logistics Infrastructure Deployment (Feb–Apr 2027)

Focus: Treat the MPM as a logistics primitive.

- Deploy swap stations at Colorado pilot sites
- Install transport frames and handling systems
- Train operators in swap-out and safety procedures
- Validate logistics repeatability (<15-minute swap target)

Milestone: *Swap-station network operational*

Phase 4 — Controlled Pilot Deployment (Apr–Aug 2027)

Focus: Real-world duty cycles.

- Run low-load and full-load mechanical tests
- Validate torque, speed, thermal, vibration envelopes
- Conduct swap-out cycles across multiple operators
- Test failure modes and emergency stop logic
- Collect telemetry for standards refinement

Milestone: *Pilot telemetry dataset complete*

Phase 5 — Standards Refinement & v1.0 Publication (Aug–Dec 2027)

Focus: Convert pilot data into standards

